Killing–Beltrami algebra and exact reductions of Navier–Stokes flow on the three-sphere

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We identify an exact $SO(4)$-equivariant interaction algebra for incompressible Navier–Stokes flow with Ebin–Marsden viscosity on the round three-sphere. The central identity determines the polarized interaction of every Killing field with every signed curl eigenspace: on $E_{k,\sigma}$, the two simple factors of the Killing algebra act by Lie derivatives with distinct chirality-dependent coefficients. This intertwining law organizes a shell-diagonal skew dynamics, a canonical generally nonflat metric connection over $SO(4)$, exact viscous finite-dimensional reductions, and a rigidity theorem stating that every finite exact reduction containing the full Killing algebra is a union of complete signed curl shells. We also obtain a quadratic-Casimir representation of the Stokes operator and viscous dissipation in terms of six Killing convective responses. Compact homogeneity then yields sharp projector norms for finite $SO(4)$-invariant vector sectors, with sharp spectral concentration consequences. The maximal torus gives a separate infinite-dimensional pressure-linear sector. These geometric and spectral statements are intrinsic to the round three-sphere and are logically independent of the conditional singularity-transfer construction that motivated the analysis.

I. INTRODUCTION AND MAIN RESULTS

Geometric formulations of incompressible fluid dynamics on Riemannian manifolds make the interaction between viscosity, isometries, and curvature structural rather than perturbative. We use the deformation-tensor (Ebin–Marsden) viscosity on the round three-sphere S^3_R, following the geometric framework of Ebin and Marsden and subsequent formulations of Navier–Stokes flow on Riemannian manifolds [3–5]. The signed curl spectrum on the round sphere is classical [6]. The geometric and spectral program from which the present work grew was developed by Batenin and Preece in a Zenodo preprint [1]; that work combines an intrinsic theory of exact spectral closure on S^3_R with a separate conditional transfer of a forced singular branch.

Two pieces of prior literature are important for positioning the results below. Samavaki and Tuomela proved that, for the same deformation-tensor viscosity on a compact Riemannian manifold, the component of an unforced solution along any Killing field is conserved [5]. The bilinear Killing cancellation used below is the polarized form of that geometric cancellation and is therefore not our principal novelty. More recently, Heslin and Preston constructed explicit nonstationary Euler solutions on curved manifolds, including the round three-sphere [7]. Thus neither conservation of Killing momentum nor the mere existence of exact Killing/eigenfield waves is the distinguishing point of the present article.

The new organizing mechanism is the exact polarized Killing–Beltrami interaction. Let

$$\mathfrak{K} = \mathfrak{K}_+ \oplus \mathfrak{K}_- \simeq \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$$

be the six-dimensional Killing algebra, with $\mathfrak{K}_\tau = E_{0,\tau}$, and let $v \in E_{k,\sigma}$ be a signed curl eigenfield. We prove

$$\boxed{2Q(K_\tau, v) = \frac{k+2-2\tau\sigma}{k+2}[K_\tau, v].} \tag{1}$$

The two simple factors therefore act on the same signed shell with the different coefficients

$$\frac{k}{k+2}, \qquad \frac{k+4}{k+2}. \tag{2}$$

This is not merely an exact-solution ansatz: it is a complete intertwining law for the polarized convective action of the Killing algebra on every signed curl shell.

Several results follow from the same identity. First, the Killing action on the orthogonal complement is shell diagonal and skew-adjoint. It transfers neither L^2 energy nor helicity between signed shells; all inter-shell nonlinear transfer is carried by the high–high term $Q(v,v)$. Second, on each $E_{k,\sigma}$ the map $m \mapsto D_{k,\sigma}(m) = 2Q(m,\cdot)$ fails in a controlled way to be a Lie-algebra representation. Its defect is exactly the curvature of a canonical metric connection on

$$SO(4) \times E_{k,\sigma} \longrightarrow SO(4),$$

and the physical Killing precession is the parallel transport of this connection. The curvature coefficients decay as k^{-1}, but the representation generators grow like k; consequently the natural operator norm of the curvature does not flatten in the ultraviolet.

Third, the intertwining formula produces finite-dimensional exact Navier–Stokes reductions

$$\mathfrak{K}_\tau \oplus E_{k,\sigma}, \qquad \mathfrak{K} \oplus E_{k,\sigma},$$

with explicit triangular viscous evolution. More importantly, we prove a converse rigidity theorem: if a finite-dimensional exact reducing space contains the full Killing algebra, then every nonzero higher spectral component is a complete signed curl shell.

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Fourth, if $T_a^\tau = Q(K_a^\tau, \cdot)$ for an L^2-orthonormal Killing basis, then

$$\boxed{A = -2V_R \sum_{\tau=\pm} \sum_{a=1}^{3} (T_a^\tau)^2,} \tag{3}$$

so the Stokes operator and viscous dissipation are represented by the quadratic Casimir of the six convective Killing responses.

Compact homogeneity supplies the complementary spectral statement

$$\|\Pi_W\|_{L^2\to L^\infty} = \sqrt{\frac{N}{3V_R}}, \qquad \|\Pi_W\|_{L^1\to L^\infty} = \frac{N}{3V_R}, \tag{4}$$

for every finite-dimensional $SO(4)$-invariant vector sector W of dimension N, with sharp constants.

The exterior-parametrix arithmetic, Mellin shift hierarchy, interaction picture, and polynomial recursion are not used in the proofs below and are outside the scope of the present article. Likewise, the conditional forced singularity transfer is not used to prove any intrinsic result here.

II. GEOMETRIC AND SPECTRAL SETTING

Let (S^3_R, g) be the round three-sphere of radius R and let

$$\begin{aligned} \partial_t u + \nabla_u u - \nu \mathcal{L}_g u + \operatorname{grad}_g p &= f, \\ \operatorname{div}_g u &= 0, \\ \mathcal{L}_g u &= 2 \operatorname{div}_g \operatorname{Def}_g u. \end{aligned}$$

Write

$$\begin{aligned} A &= -\mathbb{P}\mathcal{L}_g, \\ B(u,v) &= \mathbb{P}\nabla_u v, \\ Q(u,v) &= \frac{1}{2}\{B(u,v) + B(v,u)\}. \end{aligned}$$

Thus the projected equation is

$$\partial_t u + \nu A u + Q(u,u) = \mathbb{P} f. \tag{5}$$

The spectral conventions used here agree with Ref. [1]; in particular, the signed curl decomposition

$$L^2_\sigma(TS^3_R) = \widehat{\bigoplus}_{k\geq 0,\ \sigma=\pm} E_{k,\sigma},$$

with

$$\operatorname{curl} u = \mu_{k,\sigma} u, \qquad \mu_{k,\sigma} = \sigma \frac{k+2}{R}, \tag{6}$$

$$A u = \kappa_k u, \qquad \kappa_k = \frac{k(k+4)}{R^2}, \tag{7}$$

$$\dim E_{k,\sigma} = d_k = (k+1)(k+3). \tag{8}$$

Moreover,

$$E_{k,+} \simeq V_{(k+2)/2} \boxtimes V_{k/2}, \qquad E_{k,-} \simeq V_{k/2} \boxtimes V_{(k+2)/2} \tag{9}$$

under $SO(4) \simeq (SU(2)_+ \times SU(2)_-)/\{\pm 1\}$. The $k = 0$ shells are the two simple Killing factors:

$$\mathfrak{K} = \mathfrak{K}_+ \oplus \mathfrak{K}_-, \qquad \mathfrak{K}_\tau := E_{0,\tau}, \qquad \dim \mathfrak{K}_\tau = 3.$$

For later comparison, Ref. [1] also records

$$Q(\mathfrak{K}, \mathfrak{K}) = 0, \qquad Q(E_{k,\sigma}, E_{k,\sigma}) = 0, \tag{10}$$

and introduces the complete-shell leakage scalar

$$\mathfrak{R}(S) = \sum_{\alpha,\beta\in S} \sum_{\gamma\notin S} \left\| \Pi_\gamma Q|_{E_\alpha \times E_\beta} \right\|^2_{\mathrm{HS}}.$$

For a supplied finite union $W_S = \bigoplus_{\alpha\in S} E_\alpha$, Ref. [1] proves

$$W_S \text{ is exact reducing} \iff \mathfrak{R}(S) = 0. \tag{11}$$

We derive consequences of these intrinsic facts. No result below uses the generic-order inner solver, renormalized matching hypothesis, or annular-cycle stability hypothesis required by the relative singular transfer in Ref. [1].

The new statements use only intrinsic inputs on the round sphere: the metric, curl and Leray projection for the Killing–Beltrami interaction; the Stokes spectrum and irreducibility for Killing-saturated classification; the torus reduction and Jacobi spectrum for the maximal-torus sector; and compact homogeneity with irreducible isotropy for the projector identities.

III. THE KILLING–BELTRAMI INTERACTION ALGEBRA

We first isolate the identity that drives the new finite-dimensional closures.

Lemma III.1 (Polarized vector identity)**.** *For smooth vector fields u, v on an oriented Riemannian three-manifold,*

$$\nabla_u v + \nabla_v u = \operatorname{grad}_g\langle u, v\rangle - u \times \operatorname{curl} v - v \times \operatorname{curl} u. \tag{12}$$

If additionally $\operatorname{div}_g u = \operatorname{div}_g v = 0$*, then*

$$\operatorname{curl}(u \times v) = -[u, v]. \tag{13}$$

Proof. Both identities are tensorial versions of the standard three-dimensional vector-calculus formulas. For (12), pair both sides with an arbitrary vector field and use $\mathrm{d}(v^\flat)(X,Y) = \langle \nabla_X v, Y\rangle - \langle \nabla_Y v, X\rangle$ together with the definition $(\operatorname{curl} v)^\flat = *\,\mathrm{d}(v^\flat)$. For (13), use $(u \times v)^\flat = *(u^\flat \wedge v^\flat)$ and the Cartan identity; the divergence terms vanish, leaving $\nabla_v u - \nabla_u v = -[u, v]$. □

Lemma III.2 (Killing fields preserve curl shells)**.** *Let K be Killing and let v be divergence-free. Then*

$$\operatorname{curl}[K, v] = [K, \operatorname{curl} v]. \tag{14}$$

Consequently, if $v \in E_{k,\sigma}$, then $[K, v] \in E_{k,\sigma}$.

Proof. The flow of a Killing field consists of orientation-preserving isometries in the identity component. The Hodge star, exterior derivative, musical maps, and hence curl are natural under such isometries. Differentiating that equivariance at the identity gives (14). If $\operatorname{curl} v = \mu_{k,\sigma} v$, then $\operatorname{curl}[K, v] = \mu_{k,\sigma}[K, v]$. □

Theorem III.3 (Exact Killing–Beltrami polarization)**.** *Let*

$$K_\tau \in \mathfrak{K}_\tau, \qquad v \in E_{k,\sigma}, \qquad \tau, \sigma \in \{\pm 1\}.$$

Then

$$\boxed{Q(K_\tau, v) = \frac{\mu_{k,\sigma} - \mu_{0,\tau}}{2\mu_{k,\sigma}}[K_\tau, v].} \tag{15}$$

Equivalently,

$$\boxed{2Q(K_\tau, v) = \frac{k + 2 - 2\tau\sigma}{k+2}[K_\tau, v].} \tag{16}$$

In particular,

$$2Q(K_\sigma, v) = \frac{k}{k+2}[K_\sigma, v], \tag{17}$$

$$2Q(K_{-\sigma}, v) = \frac{k+4}{k+2}[K_{-\sigma}, v]. \tag{18}$$

Proof. Set

$$\alpha = \mu_{0,\tau} = \frac{2\tau}{R}, \qquad \beta = \mu_{k,\sigma} = \frac{\sigma(k+2)}{R}.$$

By (12),

$$\nabla_{K_\tau} v + \nabla_v K_\tau = \operatorname{grad}_g \langle K_\tau, v\rangle + (\alpha - \beta) K_\tau \times v.$$

After Leray projection,

$$Q(K_\tau, v) = \frac{\alpha - \beta}{2}\mathbb{P}(K_\tau \times v). \tag{19}$$

Since curl annihilates gradients,

$$\operatorname{curl}\mathbb{P}(K_\tau \times v) = \operatorname{curl}(K_\tau \times v) = -[K_\tau, v].$$

By the previous lemma, $[K_\tau, v]$ belongs to the β-curl eigenspace. Since $H^1(S^3) = 0$, curl has no kernel on the divergence-free subspace, so

$$\mathbb{P}(K_\tau \times v) = -\frac{1}{\beta}[K_\tau, v].$$

Substitution into (19) gives

$$Q(K_\tau, v) = \frac{\beta - \alpha}{2\beta}[K_\tau, v],$$

which is (15). Inserting the explicit eigenvalues gives (16)–(18). □

Remark III.4. The two chiralities are dynamically inequivalent once $k \geq 1$: the Killing factor with the same curl sign acts with coefficient $k/(k+2)$, whereas the opposite factor acts with coefficient $(k+4)/(k+2)$. This asymmetry is what later allows the full Killing algebra to separate the two signed shells even though the Stokes operator assigns them the same eigenvalue.

IV. EXACT MODAL COEFFICIENTS WITH ONE KILLING INPUT

The interaction formula determines not only the target shell but also the Hilbert–Schmidt coefficient used in Ref. [1].

Let

$$V_R := \operatorname{Vol}(S^3_R) = 2\pi^2 R^3.$$

Choose a pointwise orthonormal Killing frame X_1, X_2, X_3 in one simple factor $\mathfrak{K}_\tau$; then $\|X_a\|^2_{L^2} = V_R$ and, up to orientation,

$$[X_a, X_b] = \frac{2}{R}\varepsilon_{abc} X_c.$$

Thus $K_a := V_R^{-1/2} X_a$ is L^2-orthonormal in $\mathfrak{K}_\tau$.

For the action of $\mathfrak{K}_\tau$ on $E_{k,\sigma}$ define

$$j_\tau(k, \sigma) := \frac{k + 1 + \tau\sigma}{2}. \tag{20}$$

This is exactly the appropriate $SU(2)_\tau$ spin from (9).

Theorem IV.1 (Closed formula for the Killing–shell structure coefficient)**.** *Let*

$$C^{(m,\rho)}_{(0,\tau),(k,\sigma)} = \left\| \Pi_{m,\rho} Q\big|_{\mathfrak{K}_\tau \times E_{k,\sigma}} \right\|^2_{\mathrm{HS}}.$$

Then

$$C^{(m,\rho)}_{(0,\tau),(k,\sigma)} = 0 \qquad \textit{unless } (m,\rho) = (k,\sigma), \tag{21}$$

and

$$\boxed{C^{(k,\sigma)}_{(0,\tau),(k,\sigma)} = \frac{d_k}{R^2 V_R}\frac{(k+2-2\tau\sigma)^2}{(k+2)^2} \times j_\tau(k,\sigma)\big(j_\tau(k,\sigma)+1\big).} \tag{22}$$

In particular this coefficient vanishes for all $k = 0$, consistently with $Q(\mathfrak{K}, \mathfrak{K}) = 0$.

Proof. The target statement (21) follows from Theorem III.3. Put

$$c_{k,\sigma,\tau} = \frac{k + 2 - 2\tau\sigma}{2(k+2)}.$$

Then $Q(K_a, v) = c_{k,\sigma,\tau}[K_a, v]$. On the spin-$j = j_\tau(k,\sigma)$ representation, the normalized Casimir identity gives

$$-\sum_{a=1}^{3}[X_a, [X_a, \cdot]] = \frac{4}{R^2} j(j+1)\mathrm{Id}.$$

The Lie-derivative action of each X_a is skew-adjoint in L^2 because X_a is Killing. If $\{v_i\}_{i=1}^{d_k}$ is an L^2-orthonormal basis of $E_{k,\sigma}$, then

$$\sum_{a,i} \|[X_a, v_i]\|_2^2 = \frac{4}{R^2} j(j+1) d_k.$$

Replacing X_a by $K_a = X_a/\sqrt{V_R}$ and multiplying by $c_{k,\sigma,\tau}^2$ yields (22). □

Corollary IV.2 (Two explicit chiral values)**.** *For $k \geq 1$,*

$$C^{(k,\sigma)}_{(0,\sigma),(k,\sigma)} = \frac{d_k}{R^2 V_R} \frac{k^2(k+4)}{4(k+2)}, \tag{23}$$

$$C^{(k,\sigma)}_{(0,-\sigma),(k,\sigma)} = \frac{d_k}{R^2 V_R} \frac{k(k+4)^2}{4(k+2)}. \tag{24}$$

V. KILLING MOMENTUM AND SHELLWISE SKEW DYNAMICS

Samavaki and Tuomela proved conservation of the Killing component for the deformation-tensor Navier–Stokes equation [5]. The identity below is its polarized bilinear form; we include it because it couples the exact shell algebra to the full phase space.

Theorem V.1 (Polarized Killing cancellation)**.** *For every pair of smooth divergence-free vector fields u, v on S_R^3,*

$$\boxed{\Pi_{\mathfrak{K}} Q(u, v) = 0.} \tag{25}$$

Equivalently, every Killing field $K \in \mathfrak{K}$ satisfies

$$\langle Q(u, v), K \rangle_{L^2} = 0. \tag{26}$$

Proof. Since u and v are divergence-free and S_R^3 is closed, integration by parts gives

$$\langle \nabla_u v, K \rangle_{L^2} = -\int_{S_R^3} \langle v, \nabla_u K \rangle \, \mathrm{d}V,$$

and similarly with u, v interchanged. Therefore

$$2\langle Q(u, v), K \rangle_{L^2} = -\int_{S_R^3} \Big(\langle v, \nabla_u K \rangle + \langle u, \nabla_v K \rangle \Big) \, \mathrm{d}V.$$

The integrand vanishes pointwise by the Killing equation

$$\langle \nabla_X K, Y \rangle + \langle \nabla_Y K, X \rangle = 0.$$

Thus (26) holds for every $K \in \mathfrak{K}$, proving (25). □

Corollary V.2 (Exact evolution of Killing momentum)**.** *Let u solve the projected forced Navier–Stokes equation (5), and put*

$$m(t) = \Pi_{\mathfrak{K}} u(t), \qquad v(t) = (I - \Pi_{\mathfrak{K}}) u(t).$$

Then

$$\boxed{\dot{m}(t) = \Pi_{\mathfrak{K}} F(t).} \tag{27}$$

In particular, for unforced flow the full six-dimensional Killing component is conserved exactly.

The orthogonal component satisfies

$$\boxed{\partial_t v + \nu A v + Q(v, v) + D(m(t)) v = F_\perp,} \tag{28}$$

where $F_\perp = (I - \Pi_{\mathfrak{K}}) F$ and

$$D(m) v := 2Q(m, v). \tag{29}$$

If

$$v = \sum_{k \geq 1,\ \sigma = \pm} v_{k,\sigma}, \qquad m = m_+ + m_-,$$

then the action is the shell-diagonal operator

$$\boxed{D(m) v = \sum_{k \geq 1,\ \sigma = \pm} \left[\frac{k}{k+2} [m_\sigma, v_{k,\sigma}] + \frac{k+4}{k+2} [m_{-\sigma}, v_{k,\sigma}] \right].} \tag{30}$$

Proof. The Stokes operator is self-adjoint, $\ker A = \mathfrak{K}$, hence $\Pi_{\mathfrak{K}} A u = 0$. Projecting (5) to $\mathfrak{K}$ and applying Theorem V.1 gives (27). Expanding $Q(m + v, m + v)$, using $Q(m, m) = 0$ and Theorem V.1, gives (28). Formula (30) follows shell by shell from Theorem III.3. □

Remark V.3 (No single Killing drift represents the full operator). The coefficients in (30) depend on (k, σ). Consequently there is in general no single Killing field $Y_m \in \mathfrak{K}$ such that $D(m) v = [Y_m, v]$ simultaneously on every signed shell. The correct global object is the shell-diagonal skew operator $D(m)$.

If $\Pi_{\mathfrak{K}} F \in L^1([0, T); \mathfrak{K})$, then $m(t)$ remains bounded up to T. Under that assumption any L^∞ singularity must therefore occur in $v(t) \in \mathfrak{K}^\perp$.

Corollary V.4 (Affine Killing foliation)**.** *Assume $\Pi_{\mathfrak{K}} F = 0$. For every $m_0 \in \mathfrak{K}$, the affine leaf*

$$\boxed{\mathcal{H}_{m_0} := m_0 + \mathfrak{K}^\perp} \tag{31}$$

is invariant under the full projected Navier–Stokes flow. More generally, for arbitrary forcing,

$$m(t) = m_0 + \int_0^t \Pi_{\mathfrak{K}} F(s) \, \mathrm{d}s, \tag{32}$$

and the flow carries the initial leaf $\mathcal{H}_{m_0}$ into the translated leaf $\mathcal{H}_{m(t)}$.

Proof. This is exactly (27). If the Killing component of the force vanishes, $m(t) = m_0$; otherwise its evolution is the explicit affine translation (32). □

Remark V.5 (Killing-momentum fibers are not Marsden–Weinstein reduced spaces). Define the linear Killing-momentum map

$$\mathcal{J} : L^2_\sigma(TS^3_R) \longrightarrow \mathfrak{K}^*, \qquad \langle \mathcal{J}(u), K\rangle := \langle u, K\rangle_{L^2}. \quad (33)$$

After the L^2 identification $\mathfrak{K}^* \simeq \mathfrak{K}$, $\mathcal{J}(u) = \Pi_{\mathfrak{K}} u$, and therefore

$$\mathcal{J}^{-1}(m) = m + \mathfrak{K}^\perp. \quad (34)$$

Thus the affine leaves are exactly level sets of six conserved linear functionals when $\Pi_{\mathfrak{K}} F = 0$.

This statement should not be confused with Marsden–Weinstein symplectic reduction [2]. Classical reduction starts from a symplectic Hamiltonian phase space and produces the quotient $J^{-1}(\mu)/H_\mu$ by the stabilizer of the momentum value. The viscous Navier–Stokes equation is dissipative, and the affine set $\mathcal{J}^{-1}(m)$ is not itself such a quotient. In the inviscid Euler limit the conserved Killing quantities admit the usual Hamiltonian/Noether interpretation, but no symplectic-reduction claim is needed for Theorem V.4.

Theorem V.6 (Shellwise unitary Killing precession)**.** *For every $m \in \mathfrak{K}$, the operator $D(m) = 2Q(m,\cdot)$ has the following properties on the smooth divergence-free spectral core:*

$$D(m)E_{k,\sigma} \subset E_{k,\sigma}, \quad (35)$$

$$[D(m), A] = 0, \qquad [D(m), \mathrm{curl}] = 0, \quad (36)$$

$$D(m)^* = -D(m). \quad (37)$$

Consequently the nonautonomous propagator $U_m(t,s)$ solving

$$\partial_t U_m(t,s) + D(m(t))U_m(t,s) = 0, \qquad U_m(s,s) = \mathrm{Id},$$

is unitary on L^2, preserves every signed curl shell, and preserves every spectral Sobolev norm defined by a function of A.

If v solves (28), then each signed-shell energy satisfies

$$\boxed{\begin{aligned}\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|v_{k,\sigma}\|_2^2 + \nu\kappa_k\|v_{k,\sigma}\|_2^2 \\ + \langle \Pi_{k,\sigma} Q(v,v), v_{k,\sigma}\rangle \\ = \langle F_{k,\sigma}, v_{k,\sigma}\rangle\,.\end{aligned}} \quad (38)$$

In particular, the Killing component produces no transfer of energy between signed curl shells; all inter-shell nonlinear energy transfer is carried by $Q(v,v)$.

Proof. The shell invariance (35) is formula (30). Since A and curl are scalar on each signed shell, (36) follows. On each shell, (30) is a real linear combination of Lie derivatives $[K,\cdot]$ generated by Killing fields. Such Lie derivatives are skew-adjoint in L^2, proving (37). Standard finite-shell approximation and closure then give a unitary propagator preserving each spectral subspace and every A-Sobolev norm.

Project (28) onto $E_{k,\sigma}$ and take the L^2 inner product with $v_{k,\sigma}$. The Killing-drift contribution vanishes by skew-adjointness, while $Av_{k,\sigma} = \kappa_k v_{k,\sigma}$, yielding (38). □

Corollary V.7 (High-mode energy balance independent of Killing precession)**.** *For sufficiently regular solutions,*

$$\boxed{\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|v\|_2^2 + \nu\langle Av, v\rangle = \langle F_\perp, v\rangle.} \quad (39)$$

The conserved or forced Killing momentum can rotate high modes inside their shells but contributes neither positive nor negative work to the high-mode energy.

Proof. Sum (38) over all signed shells and use the usual Euler cancellation $\langle Q(v,v), v\rangle = 0$. □

VI. NONABELIAN SHELL CONNECTION AND CURVATURE

Fix a signed shell $E_{k,\sigma}$ with $k \geq 1$ and write

$$a_k = \frac{k}{k+2}, \qquad b_k = \frac{k+4}{k+2}.$$

For $m = m_\sigma + m_{-\sigma} \in \mathfrak{K}$,

$$D_{k,\sigma}(m) = a_k\ \mathrm{ad}_{m_\sigma} + b_k\ \mathrm{ad}_{m_{-\sigma}} \quad \text{on } E_{k,\sigma}.$$

Theorem VI.1 (Exact Lie-algebra representation defect)**.** *For $m, n \in \mathfrak{K}$,*

$$\begin{aligned}[D_{k,\sigma}(m), D_{k,\sigma}(n)] &= a_k^2\,\mathrm{ad}_{[m_\sigma, n_\sigma]} \\ &\quad + b_k^2\,\mathrm{ad}_{[m_{-\sigma}, n_{-\sigma}]}, \quad (40)\\ \Omega_{k,\sigma}(m,n) &:= [D_{k,\sigma}(m), D_{k,\sigma}(n)] - D_{k,\sigma}([m,n]) \\ &= -\frac{2k}{(k+2)^2}\,\mathrm{ad}_{[m_\sigma, n_\sigma]} \\ &\quad + \frac{2(k+4)}{(k+2)^2}\,\mathrm{ad}_{[m_{-\sigma}, n_{-\sigma}]}\,. \quad (41)\end{aligned}$$

Thus $m \mapsto D_{k,\sigma}(m)$ is generally not a representation of $\mathfrak{so}(4)$.

Proof. The two simple Killing factors commute: $[\mathfrak{K}_+, \mathfrak{K}_-] = 0$, and their induced Lie-derivative actions commute on $E_{k,\sigma}$. Therefore

$$\begin{aligned}&[a_k\,\mathrm{ad}_{m_\sigma} + b_k\,\mathrm{ad}_{m_{-\sigma}}, a_k\,\mathrm{ad}_{n_\sigma} + b_k\,\mathrm{ad}_{n_{-\sigma}}] \\ &\quad = a_k^2\,\mathrm{ad}_{[m_\sigma, n_\sigma]} + b_k^2\,\mathrm{ad}_{[m_{-\sigma}, n_{-\sigma}]}\,.\end{aligned}$$

Subtract

$$D_{k,\sigma}([m,n]) = a_k\,\mathrm{ad}_{[m_\sigma, n_\sigma]} + b_k\,\mathrm{ad}_{[m_{-\sigma}, n_{-\sigma}]},$$

and use $a_k(a_k - 1) = -2k/(k+2)^2$ and $b_k(b_k - 1) = 2(k+4)/(k+2)^2$. □

The defect has a precise geometric realization. Let $G = SO(4)$ and consider the trivial Hermitian bundle

$$\mathscr{E}_{k,\sigma} = G \times E_{k,\sigma} \longrightarrow G.$$

For a left-invariant vector field X^L generated by $X \in \mathfrak{so}(4) \simeq \mathfrak{K}$, define

$$\nabla^{k,\sigma}_{X^L} := X^L + D_{k,\sigma}(X). \quad (42)$$

Theorem VI.2 (Curvature of the shell-precession connection)**.** *The connection (42) is metric and its curvature on left-invariant directions is exactly*

$$\boxed{R^{k,\sigma}(X^L, Y^L) = \Omega_{k,\sigma}(X, Y).} \tag{43}$$

If $g(t) \in SO(4)$ has left-trivialized velocity

$$(L_{g(t)^{-1}})_* \dot{g}(t) = m(t),$$

then parallel transport in $\mathscr{E}_{k,\sigma}$ is exactly the Killing-precession equation

$$\dot{U}(t) + D_{k,\sigma}(m(t))U(t) = 0. \tag{44}$$

Consequently, if $g(t)$ is a closed loop, the shell evolution is the nonabelian holonomy of this connection.

Proof. Metricity follows from the skew-adjointness of every $D_{k,\sigma}(X)$. Since the coefficients are left-invariant,

$$\begin{aligned} R(X^L, Y^L) &= [\nabla_{X^L}, \nabla_{Y^L}] - \nabla_{[X^L, Y^L]} \\ &= [D(X), D(Y)] - D([X, Y]). \end{aligned}$$

which is (43). The parallel-transport equation for a connection written in left trivialization is precisely (44). □

Remark VI.3 (What holonomy does and does not say). For a sufficiently small group loop spanning directions X, Y, the leading nontrivial holonomy term is proportional to the curvature $\Omega_{k,\sigma}(X, Y)$, with the sign fixed by loop orientation and the parallel-transport convention. For a finite loop, the holonomy is a path-ordered exponential; it is not in general $\exp(\int \Omega)$.

Likewise, periodicity of the Lie-algebra-valued function $m(t)$ alone does not make the reconstructed $SO(4)$ path $g(t)$ closed. Thus the curvature theorem gives an exact geometric interpretation of shell precession but not, without an additional averaging argument, a theorem that rapid Killing forcing suppresses nonlinear triadic resonance.

Theorem VI.4 (Coefficient flattening without operator-norm flattening)**.** *Algebraically,*

$$D_{k,\sigma}(m) = \mathrm{ad}_m + \frac{2}{k+2}\left(\mathrm{ad}_{m_{-\sigma}} - \mathrm{ad}_{m_\sigma}\right). \tag{45}$$

Thus the scalar coefficients approach those of the ordinary adjoint action. However, the natural operator norms on the changing representation spaces do not flatten.

Let $X_1^\tau, X_2^\tau, X_3^\tau$ be a pointwise-orthonormal Killing frame in $\mathfrak{K}_\tau$, normalized by

$$[X_a^\tau, X_b^\tau] = \frac{2}{R}\varepsilon_{abc} X_c^\tau.$$

On $E_{k,\sigma}$ the $SU(2)_\tau$ spin is $j_\tau = (k + 1 + \tau\sigma)/2$, so

$$\| \mathrm{ad}_{X_3^\tau} \|_{\mathrm{op}, E_{k,\sigma}} = \frac{2j_\tau}{R}. \tag{46}$$

Consequently, for a unit generator in the same chiral factor,

$$\boxed{\| D_{k,\sigma}(X^\sigma) - \mathrm{ad}_{X^\sigma} \|_{\mathrm{op}} = \frac{2}{R},} \tag{47}$$

for every $k \geq 1$. For a unit generator in the opposite factor,

$$\| D_{k,\sigma}(X^{-\sigma}) - \mathrm{ad}_{X^{-\sigma}} \|_{\mathrm{op}} = \frac{2k}{(k+2)R} \longrightarrow \frac{2}{R}. \tag{48}$$

Likewise, for pointwise-orthonormal X_1^τ, X_2^τ in one simple factor, the curvature satisfies

$$\|\Omega_{k,\sigma}(X_1^\sigma, X_2^\sigma)\|_{\mathrm{op}} = \frac{4k}{(k+2)R^2} \longrightarrow \frac{4}{R^2}, \tag{49}$$

$$\|\Omega_{k,\sigma}(X_1^{-\sigma}, X_2^{-\sigma})\|_{\mathrm{op}} = \frac{4k(k+4)}{(k+2)^2 R^2} \longrightarrow \frac{4}{R^2}. \tag{50}$$

Thus the $O(k^{-1})$ curvature coefficients are exactly compensated by the $O(k)$ growth of the representation generators. The shell connection is not asymptotically flat in this natural operator norm.

Proof. In the spin-j representation of the Lie algebra normalized by $[X_a, X_b] = (2/R)\varepsilon_{abc}X_c$, a Cartan generator has eigenvalues $2im/R$, $m = -j, \ldots, j$. Hence (46). Substitution into (45) gives (47)–(48).

For curvature, take X_1^τ, X_2^τ from one simple factor. Then $[X_1^\tau, X_2^\tau] = (2/R)X_3^\tau$. Combining (41) with (46), using $j_\sigma = (k+2)/2$ in the same chiral factor and $j_{-\sigma} = k/2$ in the opposite factor, gives (49)–(50). □

Remark VI.5. A coefficient-level or suitably rescaled semiclassical curvature does decay like k^{-1}. This is distinct from operator-norm flatness on $E_{k,\sigma}$ and should not be interpreted as rigid-body convergence of the unscaled shell dynamics.

VII. EXACT KILLING–BELTRAMI REDUCTIONS

Theorem VII.1 (Killing plus one signed shell)**.** *For every $k \geq 0$ and $\sigma, \tau \in \{\pm 1\}$,*

$$\mathfrak{K}_\tau \oplus E_{k,\sigma} \tag{51}$$

is an exact Navier–Stokes reducing space. Likewise,

$$\boxed{\mathfrak{K} \oplus E_{k,\sigma} = E_{0,+} \oplus E_{0,-} \oplus E_{k,\sigma}} \tag{52}$$

is exact reducing. For $k \geq 1$ these are genuinely multi-shell closures.

Proof. The Stokes operator preserves each summand. By (10),

$$Q(\mathfrak{K}_\tau, \mathfrak{K}_\tau) = 0, \qquad Q(\mathfrak{K}, \mathfrak{K}) = 0, \qquad Q(E_{k,\sigma}, E_{k,\sigma}) = 0.$$

By Theorem III.3,

$$Q(\mathfrak{K}_\tau, E_{k,\sigma}) \subset E_{k,\sigma}, \qquad Q(\mathfrak{K}, E_{k,\sigma}) \subset E_{k,\sigma}.$$

Therefore the polarized quadratic term has no leakage from either space. □

Corollary VII.2 (New zeros of the complete-shell leakage scalar)**.** *Let*

$$S_{\tau;k,\sigma} := \{(0,\tau),(k,\sigma)\},$$
$$S_{K;k,\sigma} := \{(0,+),(0,-),(k,\sigma)\}.$$

Then

$$\mathfrak{R}(S_{\tau;k,\sigma}) = 0, \qquad \boxed{\mathfrak{R}(S_{K;k,\sigma}) = 0.}$$

VIII. EXACT VISCOUS NONSTATIONARY SOLUTIONS

Fix $k \geq 1$ and $\sigma \in \{\pm 1\}$. Decompose a Killing field as

$$K = K_\sigma + K_{-\sigma}, \qquad K_{\pm\sigma} \in \mathfrak{K}_{\pm\sigma}.$$

Define

$$D_{k,\sigma}(K)v := \frac{k}{k+2}[K_\sigma, v] + \frac{k+4}{k+2}[K_{-\sigma}, v]. \tag{53}$$

By Theorem III.3,

$$D_{k,\sigma}(K)v = 2Q(K,v). \tag{54}$$

Notice that

$$D_{k,\sigma}(K) = \mathrm{ad}_{\widetilde{K}}, \qquad \widetilde{K} = \frac{k}{k+2}K_\sigma + \frac{k+4}{k+2}K_{-\sigma},$$

so $D_{k,\sigma}(K)$ is skew-adjoint on L^2.

Theorem VIII.1 (Triangular exact dynamics)**.** *Let*

$$u(t) = K(t) + v(t), \qquad K(t) \in \mathfrak{K}, \quad v(t) \in E_{k,\sigma},$$

and let the projected force be $F = F_{\mathfrak{K}} + F_k$ in the same space. Then (5) is exactly equivalent to

$$\dot{K} = F_{\mathfrak{K}}, \tag{55}$$
$$\dot{v} + \nu\kappa_k v + D_{k,\sigma}(K)v = F_k. \tag{56}$$

Thus the original quadratic PDE becomes a triangular finite-dimensional system: the Killing component is independent and the shell component is linear once $K(t)$ is known.

Proof. Expand

$$Q(K+v, K+v) = Q(K,K) + 2Q(K,v) + Q(v,v).$$

The first and last terms vanish by (10); the middle term is (54). Also $AK = 0$ and $Av = \kappa_k v$. Orthogonality of the Stokes eigenspaces gives (55)–(56). □

Corollary VIII.2 (Closed unforced formula)**.** *For unforced data $K(0) = K_0$, $v(0) = v_0$,*

$$\boxed{u(t) = K_0 + e^{-\nu k(k+4)t/R^2} e^{-tD_{k,\sigma}(K_0)} v_0.} \tag{57}$$

Moreover,

$$\boxed{\|v(t)\|_2 = e^{-\nu k(k+4)t/R^2}\|v_0\|_2.} \tag{58}$$

In the inviscid limit the shell component evolves by a one-parameter isometry representation and has constant L^2 norm.

Proof. In the unforced case $K(t) = K_0$. Since $D_{k,\sigma}(K_0)$ is time-independent and skew-adjoint, (56) integrates to (57); skew-adjointness gives (58). □

Remark VIII.3 (Relation to exact Euler waves). Recent work of Heslin and Preston constructs exact nonstationary Euler solutions on curved manifolds, including the round three-sphere, by perturbing rigid motions with suitable eigenfields [7]. Exact Navier–Stokes solutions generated by Beltrami structures have also been studied in Euclidean settings [8]. Formula (57) is a viscous exact closure for the Ebin–Marsden viscosity used here. The present derivation is internal to the Preece–Batenin spectral data and does not require importing the Heslin–Preston construction.

IX. RIGIDITY OF KILLING-SATURATED REDUCTIONS

The previous theorem gives examples. We now prove a converse that changes the status of the complete-shell criterion in an important subclass.

Theorem IX.1 (Shell completeness forced by the full Killing algebra)**.** *Let W be a finite-dimensional exact Navier–Stokes reducing space of smooth divergence-free vector fields on S^3_R, and assume*

$$\mathfrak{K} = E_{0,+} \oplus E_{0,-} \subset W. \tag{59}$$

Then there exists a finite set $S \subset \mathbb{N}_0 \times \{\pm\}$ containing $(0,+)$ and $(0,-)$ such that

$$\boxed{W = \bigoplus_{(k,\sigma)\in S} E_{k,\sigma}.} \tag{60}$$

That is, every nonzero higher component is forced to be a complete signed curl shell.

Proof. Because W is reducing, $AW \subset W$. The operator A is self-adjoint and has the distinct eigenvalues $\kappa_k = k(k+4)/R^2$ indexed by k; the κ_k eigenspace is

$$F_k := E_{k,+} \oplus E_{k,-}.$$

Since W is finite-dimensional, spectral interpolation in the finite set of eigenvalues appearing in W yields

$$W = \bigoplus_k W_k, \qquad W_k := W \cap F_k.$$

By assumption $W_0 = F_0 = \mathfrak{K}$.

Fix $k \geq 1$ with $W_k \neq 0$. Put

$$a = \frac{k}{k+2}, \qquad b = \frac{k+4}{k+2};$$

then a, b are nonzero and $a \neq b$.

Take $X \in \mathfrak{K}_+$. Since $\mathfrak{K}_+ \subset W$ and W is quadratically closed, the operator

$$D_X := 2Q(X, \cdot)$$

preserves W_k. By Theorem III.3,

$$D_X = a\,\rho_+(X) \oplus b\,\rho_-(X) \quad \text{on } F_k = E_{k,+} \oplus E_{k,-}, \tag{61}$$

where $\rho_\pm(X) = [X, \cdot]$ on $E_{k,\pm}$.

If $X, Y \in \mathfrak{K}_+$ and $Z = [X, Y]$, then $[D_X, D_Y]$ also preserves W_k and

$$[D_X, D_Y] = a^2\rho_+(Z) \oplus b^2\rho_-(Z).$$

Since $\mathfrak{su}(2)$ is perfect, every $Z \in \mathfrak{K}_+$ is a linear combination of commutators. Linear combinations of D_Z and the preceding commutator isolate

$$\rho_+(Z) \oplus 0, \qquad 0 \oplus \rho_-(Z),$$

because the coefficient matrix $\left(\begin{smallmatrix} a & b \\ a^2 & b^2 \end{smallmatrix}\right)$ is invertible.

Hence W_k is invariant under the block-supported $\mathfrak{K}_+$ actions. The quadratic Casimir of the first block equals a positive scalar times the orthogonal projector onto $E_{k,+}$ and vanishes on $E_{k,-}$; similarly for the second block. Therefore

$$W_k = (W_k \cap E_{k,+}) \oplus (W_k \cap E_{k,-}).$$

Call these intersections $L_{k,+}$ and $L_{k,-}$.

Now use the $\mathfrak{K}_-$ part of the contained Killing algebra. By Theorem III.3, its action on each $E_{k,\sigma}$ again has a nonzero coefficient. Therefore each $L_{k,\sigma}$ is invariant under both simple factors $\mathfrak{K}_+$ and $\mathfrak{K}_-$. Thus $L_{k,\sigma}$ is an $SO(4)$-submodule of $E_{k,\sigma}$. By irreducibility (9),

$$L_{k,\sigma} = 0 \quad \text{or} \quad L_{k,\sigma} = E_{k,\sigma}.$$

Doing this for every k proves (60). □

Corollary IX.2 (Complete classification of Killing-saturated finite reductions)**.** *A finite-dimensional smooth subspace W satisfying $\mathfrak{K} \subset W$ is an exact reducing space if and only if*

$$W = W_S = \bigoplus_{\alpha \in S} E_\alpha$$

for a finite shell set S containing $(0,+), (0,-)$ and satisfying

$$\boxed{\mathfrak{R}(S) = 0.} \tag{62}$$

Thus, inside the Killing-saturated category, the complete-shell leakage criterion is both a test and an exhaustive classification.

Proof. The forward implication is Theorem IX.1 followed by the criterion of Ref. [1] (11). The reverse implication is exactly (11). □

Corollary IX.3 (One-shell rigidity)**.** *Fix $k \geq 1$ and σ. If*

$$\mathfrak{K} \subset W \subset \mathfrak{K} \oplus E_{k,\sigma}$$

and W is exact reducing, then

$$W = \mathfrak{K} \quad \text{or} \quad W = \mathfrak{K} \oplus E_{k,\sigma}.$$

X. CASIMIR REPRESENTATION OF STOKES DISSIPATION

Let X_a^+, X_a^-, $a = 1, 2, 3$, be the standard left- and right-invariant global Killing frames on the round three-sphere, normalized so that each triple is pointwise orthonormal. Set

$$V_R = \mathrm{Vol}(S_R^3), \qquad K_a^\pm = V_R^{-1/2} X_a^\pm.$$

Then the six $K_a^\pm$ are L^2-orthonormal in $\mathfrak{K}$. Define

$$T_a^\pm := Q(K_a^\pm, \cdot).$$

Theorem X.1 (Exact quadratic Casimir identity)**.** *On smooth divergence-free vector fields,*

$$\boxed{A = -2V_R \sum_{\tau=\pm} \sum_{a=1}^{3} (T_a^\tau)^2.} \tag{63}$$

Equivalently, for every $u \in H^1_\sigma(TS_R^3)$,

$$\boxed{\langle Au, u \rangle_{L^2} = 2V_R \sum_{\tau=\pm} \sum_{a=1}^{3} \|Q(K_a^\tau, u)\|_2^2.} \tag{64}$$

Proof. It is enough to work on a signed shell $E_{k,\sigma}$ and then sum orthogonally. By Theorem IV.1, the Hilbert–Schmidt trace per unit vector in the shell is

$$\sum_{\tau=\pm} \frac{C^{(k,\sigma)}_{(0,\tau),(k,\sigma)}}{d_k}.$$

Using (23)–(24),

$$\begin{aligned} \sum_{\tau=\pm} \frac{C_\tau}{d_k} &= \frac{1}{R^2 V_R} \left[\frac{k^2(k+4)}{4(k+2)} + \frac{k(k+4)^2}{4(k+2)} \right] \\ &= \frac{k(k+4)}{2R^2 V_R} = \frac{\kappa_k}{2V_R}. \end{aligned}$$

By $SO(4)$-equivariance the positive operator $\sum_{\tau,a}(T_a^\tau)^* T_a^\tau$ is scalar on the irreducible $E_{k,\sigma}$, and its scalar is therefore $\kappa_k/(2V_R)$. Hence

$$A = 2V_R \sum_{\tau,a} (T_a^\tau)^* T_a^\tau.$$

Each T_a^τ is skew-adjoint: on every signed shell it is a real scalar multiple of the Lie derivative $[K_a^\tau, \cdot]$, and Killing Lie derivatives are skew-adjoint in L^2. Thus $(T_a^\tau)^* = -T_a^\tau$, yielding (63). □

There is also a simpler relation with the ordinary isometry Casimir. Define

$$\Delta_{\mathrm{iso}} := -\sum_{\tau=\pm}\sum_{a=1}^{3} \mathcal{L}^2_{X_a^\tau}. \tag{65}$$

Theorem X.2 (Correct Stokes–isometry-Casimir relation)**.** *On divergence-free vector fields on the round three-sphere,*

$$\boxed{\Delta_{\mathrm{iso}} = 2\ \mathrm{curl}^2, \qquad A = \frac{1}{2}\Delta_{\mathrm{iso}} - \frac{4}{R^2}\mathrm{Id}.} \tag{66}$$

If instead the L^2-normalized fields K_a^τ are used to define $\Delta_K = -\sum_{\tau,a} \mathcal{L}^2_{K_a^\tau}$, then

$$\boxed{A = \frac{V_R}{2}\Delta_K - \frac{4}{R^2}\mathrm{Id}.} \tag{67}$$

Proof. On $E_{k,\sigma}$ the two $SU(2)$ spins are

$$j_+ = \frac{k+1+\sigma}{2}, \qquad j_- = \frac{k+1-\sigma}{2}.$$

With the normalization of the pointwise-unit frames, the two Casimirs have eigenvalues $4j_\pm(j_\pm+1)/R^2$. Their sum is

$$\frac{4}{R^2}\left(j_+(j_++1) + j_-(j_-+1)\right) = \frac{2(k+2)^2}{R^2}.$$

Since curl^2 has eigenvalue $(k+2)^2/R^2$ and A has eigenvalue $k(k+4)/R^2 = ((k+2)^2-4)/R^2$, the identities follow on every signed shell and hence spectrally on their direct sum. □

Corollary X.3 (Exact dissipation as six convective responses)**.** *For $u \in H^1_\sigma(TS^3_R)$,*

$$\boxed{2\nu\|\operatorname{Def} u\|_2^2 = \nu\langle Au, u\rangle = 2\nu V_R \sum_{\tau=\pm}\sum_{a=1}^{3} \|Q(K_a^\tau, u)\|_2^2.} \tag{68}$$

Consequently, if the dissipation tends to infinity along a sequence of times, then the maximum of these six response norms tends to infinity along a subsequence.

Remark X.4 (What the six-response identity does not imply). Equation (68) is an exact global L^2 identity. It does not by itself imply microlocal anisotropy, spatial polarization of a singular core, or concentration along a preferred geometric axis. The rigorous conclusion is only the finite-sum statement in the corollary: unbounded dissipation forces at least one of the six global response norms to become unbounded along a subsequence.

Proof. Integration by parts in $A = -\mathbb{P}(2\,\mathrm{div}\,\mathrm{Def})$ gives $\langle Au, u\rangle = 2\|\operatorname{Def} u\|_2^2$. Now apply (64). The final statement follows because there are only six nonnegative summands. □

XI. THE ENTIRE MAXIMAL-TORUS SECTOR IS PRESSURE-LINEAR

Reference [1] proves that every smooth $T^2 = U(1)_\theta \times U(1)_z$ invariant divergence-free field has the form

$$u = \alpha(s)\partial_\theta + \beta(s)\partial_z, \qquad s = r^2/R^2,$$

and that its nonlinear term is absorbed exactly into pressure.

Let $\mathcal{V}_{T^2}$ denote the full infinite-dimensional space of these fields.

Theorem XI.1 (Infinite-dimensional pressure-linearity)**.** *For all $u, v \in \mathcal{V}_{T^2}$,*

$$\boxed{Q(u, v) = 0.} \tag{69}$$

Hence the projected Navier–Stokes equation is exactly linear on the entire maximal-torus invariant sector.

Proof. The torus reduction of Ref. [1] gives $Q(w, w) = 0$ for every $w \in \mathcal{V}_{T^2}$. Since $\mathcal{V}_{T^2}$ is a vector space, polarization yields

$$2Q(u, v) = Q(u+v, u+v) - Q(u, u) - Q(v, v) = 0.$$

□

Reference [1] also diagonalizes the Stokes operator with Jacobi modes

$$e_n^\theta = P_n^{(1,0)}(1-2s)\partial_\theta, \tag{70}$$
$$e_n^z = P_n^{(0,1)}(1-2s)\partial_z, \tag{71}$$

and

$$H_n := \mathrm{span}\{e_n^\theta, e_n^z\}, \qquad A|_{H_n} = \lambda_n \mathrm{Id}, \qquad \lambda_n = \frac{4n(n+2)}{R^2}. \tag{72}$$

The λ_n are strictly increasing.

Theorem XI.2 (Finite toric classification)**.** *A finite-dimensional subspace $W \subset \mathcal{V}_{T^2}$ is exact reducing if and only if*

$$\boxed{W = \bigoplus_{n\in F} L_n, \qquad L_n \subset H_n,} \tag{73}$$

for some finite set $F \subset \mathbb{N}_0$ and arbitrary linear subspaces L_n of the two-dimensional eigenspaces H_n.

Proof. By Theorem XI.1, quadratic closure is automatic. Thus W is reducing if and only if $AW \subset W$. Since A is self-adjoint and the eigenvalues (72) are distinct, every finite-dimensional A-invariant subspace splits into its intersections with the eigenspaces H_n, giving (73). Conversely any such direct sum is A-invariant. □

Corollary XI.3 (Global smooth toric dynamics)**.** *For smooth T^2-invariant initial data and smooth T^2-invariant forcing, the smooth solution remains smooth for every finite time for which the force is smooth. In the unforced case,*

$$\|u(t) - \Pi_{H_0} u_0\|_{H^s} \le e^{-12\nu t/R^2} \|u_0 - \Pi_{H_0} u_0\|_{H^s} \tag{74}$$

for every spectral Sobolev index s. Thus finite-time singularity is impossible in the entire infinite-dimensional maximal-torus sector.

Proof. The sector is a pair of scalar Jacobi heat equations. The first positive eigenvalue is $\lambda_1 = 12/R^2$, while H_0 is the stationary two-dimensional Killing subspace. The spectral formula gives (74), and Duhamel's formula handles smooth forcing. □

XII. A HOMOGENEOUS ADDITION THEOREM FOR VECTOR PROJECTORS

The exact projector estimate in Ref. [1] is not special to the canonical band. The correct principle is homogeneity plus irreducible isotropy.

Theorem XII.1 (Homogeneous-bundle evaluation theorem)**.** *Let $M = G/H$ be a compact homogeneous Riemannian manifold and let $\mathcal{E} \to M$ be a rank-r real Euclidean homogeneous vector bundle whose isotropy representation on a fiber is irreducible. Let $W \subset C^\infty(\mathcal{E})$ be a finite-dimensional G-invariant subspace and let Π_W be the L^2-orthogonal projector. If $N = \dim W$ and $V = \mathrm{Vol}(M)$, then*

$$\boxed{\|\Pi_W\|_{L^2\to L^\infty} = \sqrt{\frac{N}{rV}},} \tag{75}$$

$$\boxed{\|\Pi_W\|_{L^1\to L^\infty} = \frac{N}{rV}.} \tag{76}$$

Both constants are sharp.

Proof. Let $K_W(x,y) : \mathcal{E}_y \to \mathcal{E}_x$ be the projector kernel and $E_x : W \to \mathcal{E}_x$ the evaluation map. Then

$$K_W(x,y) = E_x E_y^*, \qquad K_W(x,x) = E_x E_x^*.$$

At a fixed point x, $K_W(x,x)$ commutes with the isotropy action of H. By irreducibility and self-adjointness it is a scalar:

$$K_W(x,x) = c\,\mathrm{Id}_{\mathcal{E}_x}.$$

Transitivity makes c independent of x. Taking the operator trace,

$$N = \operatorname{tr}\Pi_W = \int_M \operatorname{tr} K_W(x,x)\,\mathrm{d}V = rcV,$$

so $c = N/(rV)$.

For $L^2 \to L^\infty$,

$$|(\Pi_W f)(x)| \le \|E_x\|\|f\|_2 = \sqrt{c}\,\|f\|_2.$$

Equality is attained by $f = E_x^*\xi/\|E_x^*\xi\|_2$ for any unit ξ.

For $L^1 \to L^\infty$,

$$\|K_W(x,y)\|_{\mathrm{op}} \le \|E_x\|\|E_y\| = c,$$

giving the upper bound. The diagonal value is exactly c; a vector-valued approximate identity concentrated at $y = x$ and polarized along an eigenvector of $K_W(x,x)$ approaches this value. Hence the bound is sharp. □

Corollary XII.2 (Every finite $SO(4)$-invariant vector sector on S^3)**.** *Let W be any finite-dimensional $SO(4)$-invariant subspace of smooth divergence-free vector fields on S^3_R. Then, with $N = \dim W$,*

$$\boxed{\|\Pi_W\|_{2\to\infty} = \sqrt{\frac{N}{3V_R}},} \tag{77}$$

$$\boxed{\|\Pi_W\|_{1\to\infty} = \frac{N}{3V_R}.} \tag{78}$$

Proof. Apply Theorem XII.1 to the tangent bundle of the round three-sphere. The isotropy group is $SO(3)$ acting irreducibly on the three-dimensional tangent space. □

XIII. SHARP SPECTRAL UNCERTAINTY AND ADAPTIVE SHELL SPARSITY

Corollary XIII.1 (Sharp L^2 concentration-volume inequality)**.** *If $0 \ne w \in W$ and W is as in Theorem XII.2, then*

$$\boxed{\frac{\|w\|_2^2}{\|w\|_\infty^2} \ge \frac{3V_R}{\dim W}.} \tag{79}$$

The constant is sharp for every such W.

Proof. The inequality is Theorem XII.2 applied to $w = \Pi_W w$. Sharpness follows from the reproducing-kernel extremizer $E_x^* \xi$ used in Theorem XII.1. □

Corollary XIII.2 (Sharp uncertainty inside one signed curl shell)**.** *For $w \in E_{k,\sigma}$,*

$$\boxed{\frac{\|w\|_2^2}{\|w\|_\infty^2} \geq \frac{3V_R}{(k+1)(k+3)}.} \tag{80}$$

Equality is attained. Any extremizer is still a curl eigenfield and hence generates the exact unforced Navier–Stokes trajectory

$$u(t) = e^{-\nu k(k+4)t/R^2} w.$$

Now let S be an arbitrary finite set of signed shell labels and put

$$W_S = \bigoplus_{(k,\sigma)\in S} E_{k,\sigma}, \qquad N_S = \sum_{(k,\sigma)\in S} (k+1)(k+3).$$

Theorem XIII.3 (Adaptive rank barrier)**.** *For any integrable divergence-free field v, if*

$$\|\Pi_{W_S} v\|_\infty \geq \eta \|v\|_\infty, \qquad 0 < \eta \leq 1, \tag{81}$$

then

$$\boxed{N_S \geq 3V_R\, \eta\, \frac{\|v\|_\infty}{\|v\|_1}.} \tag{82}$$

No contiguity or low-frequency assumption on S is required.

Proof. By Theorem XII.2,

$$\|\Pi_{W_S} v\|_\infty \leq \frac{N_S}{3V_R} \|v\|_1.$$

Combine this with (81) and rearrange. □

Theorem XIII.4 (Derivative-budget shell-count barrier)**.** *Let S be a finite set of M signed curl shells and assume $\Delta_1 v \in L^1$, where $\Delta_1 = \mathrm{curl}^2$ on coexact one-forms. Then*

$$\boxed{\|\Pi_{W_S} v\|_\infty \leq \frac{R^2}{3V_R} \left[\sum_{(k,\sigma)\in S} \left(1 - \frac{1}{(k+2)^2}\right)\right] \|\Delta_1 v\|_1.} \tag{83}$$

Hence, if $\|\Pi_{W_S} v\|_\infty \geq \eta \|v\|_\infty$, then

$$\boxed{M \geq \frac{3V_R\, \eta}{R^2} \frac{\|v\|_\infty}{\|\Delta_1 v\|_1}.} \tag{84}$$

This lower bound is independent of the maximum frequency of the selected shells.

Proof. On $E_{k,\sigma}$,

$$\Delta_1 = \frac{(k+2)^2}{R^2} \mathrm{Id}.$$

Therefore

$$\Pi_{k,\sigma} v = \frac{R^2}{(k+2)^2} \Pi_{k,\sigma} \Delta_1 v.$$

The exact $L^1 \to L^\infty$ projector norm gives

$$\begin{aligned} \|\Pi_{k,\sigma} v\|_\infty &\leq \frac{R^2}{(k+2)^2} \frac{(k+1)(k+3)}{3V_R} \|\Delta_1 v\|_1 \\ &= \frac{R^2}{3V_R} \left(1 - \frac{1}{(k+2)^2}\right) \|\Delta_1 v\|_1. \end{aligned}$$

Summing over S proves (83). Since every weight in the sum is strictly less than one, the peak-capture condition implies (84). □

Theorem XIII.5 (Fractional L^1 shell barrier)**.** *Let $s > 0$ and let $\Delta_1 = \mathrm{curl}^2$ on divergence-free one-forms. For any finite signed-shell set S,*

$$\boxed{\|\Pi_{W_S} v\|_\infty \leq \frac{R^s}{3V_R} \left[\sum_{(k,\sigma)\in S} \frac{(k+1)(k+3)}{(k+2)^s}\right] \|\Delta_1^{s/2} v\|_1.} \tag{85}$$

Consequently, if $\|\Pi_{W_S} v\|_\infty \geq \eta \|v\|_\infty$, then

$$\boxed{\sum_{(k,\sigma)\in S} \frac{(k+1)(k+3)}{(k+2)^s} \geq \frac{3V_R\, \eta}{R^s} \frac{\|v\|_\infty}{\|\Delta_1^{s/2} v\|_1}.} \tag{86}$$

For $s = 2$ this recovers the weighted estimate (83).

Proof. On $E_{k,\sigma}$,

$$\Delta_1^{s/2} = \left(\frac{k+2}{R}\right)^s \mathrm{Id}.$$

Hence

$$\Pi_{k,\sigma} v = \left(\frac{R}{k+2}\right)^s \Pi_{k,\sigma} \Delta_1^{s/2} v.$$

Using the exact projector norm

$$\|\Pi_{k,\sigma}\|_{L^1 \to L^\infty} = \frac{(k+1)(k+3)}{3V_R}$$

and summing over S gives (85). The capture inequality follows by rearrangement. □

Corollary XIII.6 (Explicit global L^1 Bessel-potential embedding for $s > 3$)**.** *If $s > 3$ and $\Delta_1^{s/2} v \in L^1$, then*

$$\boxed{\|v\|_\infty \leq \frac{2R^s}{3V_R} \big[\zeta(s-2) - \zeta(s)\big] \|\Delta_1^{s/2} v\|_1.} \tag{87}$$

Thus an L^∞ blow-up sequence with bounded $\|\Delta_1^{s/2} v\|_1$ is impossible for every $s > 3$.

Proof. Put $j = k+2$. Since

$$\frac{(k+1)(k+3)}{(k+2)^s} = \frac{j^2-1}{j^s} = j^{2-s} - j^{-s},$$

the sum over both signs and all $k \geq 0$ is

$$2\sum_{j=2}^{\infty}(j^{2-s} - j^{-s}) = 2\big[\zeta(s-2) - \zeta(s)\big],$$

which converges exactly when $s > 3$. Apply (85) to an increasing sequence of finite shell sets; the right-hand side is summable in L^∞, so the shell expansion converges absolutely there and gives (87). □

Remark XIII.7 (Interpretation of the threshold). The condition $s > 3$ is the summability threshold of this L^1 shell argument in spatial dimension three. It is an L^1 Bessel-potential statement, not the usual L^2 Sobolev condition $H^s \hookrightarrow L^\infty$. No claim of endpoint optimality at $s = 3$ is made.

Proposition XIII.8 (Exact pole of the supercritical embedding constant). *Write the constant in* (87) *as*

$$C_s(R) := \frac{2R^s}{3V_R}\big[\zeta(s-2) - \zeta(s)\big] = \frac{R^{s-3}}{3\pi^2}\big[\zeta(s-2) - \zeta(s)\big]. \tag{88}$$

Then

$$\boxed{\lim_{s\downarrow 3}(s-3)C_s(R) = \frac{1}{3\pi^2}.} \tag{89}$$

Proof. Put $s = 3+\varepsilon$. Since

$$\varepsilon\,\zeta(1+\varepsilon) \longrightarrow 1, \qquad \varepsilon\,\zeta(3+\varepsilon) \longrightarrow 0, \qquad R^\varepsilon \longrightarrow 1,$$

formula (89) follows from (88). □

Remark XIII.9 (The pole is not an endpoint Trudinger theorem). The residue (89) describes the divergence of the constants in the family of estimates for $s > 3$. It does not permit one to set $s = 3$ in (87), because both the spectral series and the norm $\|\Delta_1^{s/2}v\|_1$ vary with s. In particular no $W^{3,1} \to$ BMO embedding, no Trudinger–Moser exponential inequality, and no sharp endpoint constant follow from the zeta-pole calculation alone. Such endpoint statements require an independent harmonic-analysis theorem.

XIV. EXACT SHELL-HELICITY BALANCES

For $v_{k,\sigma} \in E_{k,\sigma}$ define its signed shell helicity

$$\begin{aligned}\mathscr{H}_{k,\sigma}(v) &:= \langle v_{k,\sigma}, \operatorname{curl} v_{k,\sigma}\rangle \\ &= \mu_{k,\sigma}\|v_{k,\sigma}\|_2^2, \qquad \mu_{k,\sigma} = \sigma\frac{k+2}{R}.\end{aligned} \tag{90}$$

Theorem XIV.1 (Killing transparency and shell-helicity balance). *For every shell,*

$$\boxed{\langle D(m)v_{k,\sigma}, \operatorname{curl} v_{k,\sigma}\rangle = 0.} \tag{91}$$

Along a sufficiently regular solution of the high-mode equation,

$$\begin{aligned}\boxed{\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{H}_{k,\sigma} + \nu\kappa_k\mathscr{H}_{k,\sigma} \\ + \mu_{k,\sigma}\,\langle \Pi_{k,\sigma}Q(v,v), v_{k,\sigma}\rangle \\ = \mu_{k,\sigma}\,\langle F_{k,\sigma}, v_{k,\sigma}\rangle\,.}\end{aligned} \tag{92}$$

Thus Killing precession transfers neither energy nor helicity between signed curl shells.

Proof. Since $D(m)$ is skew-adjoint and commutes with curl,

$$\begin{aligned}\langle Dv, \operatorname{curl} v\rangle &= -\langle v, D\operatorname{curl} v\rangle \\ &= -\langle v, \operatorname{curl} Dv\rangle \\ &= -\langle \operatorname{curl} v, Dv\rangle.\end{aligned}$$

so the real scalar equals its own negative and is zero. The balance (92) is obtained by multiplying the shell-energy identity (38) by the scalar $\mu_{k,\sigma}$. □

Theorem XIV.2 (High-mode helicity law on every affine Killing leaf). *Let*

$$\mathscr{H}(v) = \langle v, \operatorname{curl} v\rangle.$$

Then

$$\boxed{\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{H}(v) + \nu\langle Av, \operatorname{curl} v\rangle = \langle F_\perp, \operatorname{curl} v\rangle.} \tag{93}$$

In particular, for inviscid unforced dynamics on a fixed affine leaf, $\mathscr{H}(v)$ is conserved exactly.

Proof. Sum (92) over all signed shells. It remains to show that the nonlinear sum vanishes:

$$\langle Q(v,v), \operatorname{curl} v\rangle = 0.$$

Because $\operatorname{curl} v$ is divergence-free, the Leray projection can be removed. Using the three-dimensional identity

$$\nabla_v v = \operatorname{grad}_g\left(\frac{1}{2}|v|^2\right) - v \times \operatorname{curl} v,$$

the gradient term integrates to zero against $\operatorname{curl} v$, while the cross product is pointwise orthogonal to $\operatorname{curl} v$. Hence the nonlinear contribution is zero. The Killing contribution already vanishes by (91). □

Proposition XIV.3 (Signed helicity dissipation and absolute chiral damping). *The viscous term in the high-mode helicity law is*

$$\boxed{\nu\langle Av, \operatorname{curl} v\rangle = \frac{\nu}{R}\sum_{k\geq 1}\kappa_k(k+2)\left(\|v_{k,+}\|_2^2 - \|v_{k,-}\|_2^2\right).} \tag{94}$$

It is therefore sign-indefinite as a functional of a mixed-chirality field.

Nevertheless viscosity is not antidissipative on the negative-helicity sector. With nonlinear transfer and forcing suppressed, every shell obeys

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{H}_{k,\sigma}\Big|_{\nu} &= -2\nu\kappa_k\mathscr{H}_{k,\sigma},\\ \frac{\mathrm{d}}{\mathrm{d}t}|\mathscr{H}_{k,\sigma}|\Big|_{\nu} &= -2\nu\kappa_k|\mathscr{H}_{k,\sigma}|.\end{aligned} \tag{95}$$

Thus viscosity drives both positive and negative shell helicities toward zero.

If

$$\mathscr{H}_{\mathrm{abs}}(v) := \sum_{k,\sigma}|\mu_{k,\sigma}|\|v_{k,\sigma}\|_2^2, \tag{96}$$

then its purely viscous contribution satisfies

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{H}_{\mathrm{abs}}(v)\Big|_{\nu} = -2\nu\sum_{k,\sigma}\kappa_k|\mu_{k,\sigma}|\|v_{k,\sigma}\|_2^2 \le 0.} \tag{97}$$

Proof. Insert $Av_{k,\sigma} = \kappa_k v_{k,\sigma}$ and $\operatorname{curl} v_{k,\sigma} = \sigma(k+2)R^{-1}v_{k,\sigma}$ into the inner product. Equation (95) follows from the shell balance with only the viscous term retained, and summation gives (97). □

Remark XIV.4. The sign-indefiniteness of (94) reflects the signed definition of total helicity, not a viscous instability of one chirality. Likewise, a fixed sign of an azimuthal physical-space swirl does not by itself determine the spectral imbalance between $E_{k,+}$ and $E_{k,-}$. No chiral singularity-selection mechanism is inferred here.

Remark XIV.5 (No chiral selection rule is inferred). The conservation law constrains the signed sum of nonlinear shell transfers, but it does not by itself forbid direct interactions between $E_{k,+}$ and $E_{\ell,-}$. A claimed "chiral valve" would require an explicit Clebsch–Gordan analysis of the polarized map $Q(E_{k,+}, E_{\ell,-})$ and its target representations. No such additional selection rule is assumed here.

XV. SHARP LIMITATIONS OF STRONGER CLASSIFICATION AND ENTROPY CLAIMS

Theorem XV.1 (Grassmannian families of exact pressure-linear reductions)**.** *Fix a signed curl shell $E_{k,\sigma}$. Every linear subspace $L \subset E_{k,\sigma}$ is an exact reducing space:*

$$AL \subset L, \qquad Q(L,L) = 0. \tag{98}$$

Consequently, for every $1 \le d \le d_k = (k+1)(k+3)$, the whole Grassmannian

$$\mathrm{Gr}(d, E_{k,\sigma}) \tag{99}$$

consists of d-dimensional exact Navier–Stokes reductions.

Likewise every subspace $H \subset \mathfrak{K}$ is exact, because $A|_{\mathfrak{K}} = 0$ and $Q(\mathfrak{K},\mathfrak{K}) = 0$.

Proof. The Stokes operator is scalar on $E_{k,\sigma}$: $A = \kappa_k\mathrm{Id}$. Moreover $Q(E_{k,\sigma}, E_{k,\sigma}) = 0$ by the Beltrami cancellation and polarization. Both assertions therefore restrict to every subspace $L \subset E_{k,\sigma}$. The Killing statement is identical with $\kappa_0 = 0$ and the pressure-linearity of the full Killing algebra. □

Corollary XV.2 (No finite atlas of all reductions in dimension at most eight)**.** *For every $1 \le d \le 8$ there are continuous families of exact d-dimensional reducing spaces on S^3_R. Hence no finite list of isometry types of the form "Killing, toric, one shell, resonant sector" can be an exhaustive classification of all exact reductions with $\dim W \le 8$ without additional hypotheses.*

Proof. Choose k with $d_k \ge d$ and apply Theorem XV.1. The Grassmannian $\mathrm{Gr}(d, E_{k,\sigma})$ has positive dimension whenever $0 < d < d_k$; for $d = 8$ one may take any $k \ge 2$, for which $d_k > 8$. □

Proposition XV.3 (The nonlinear-coherence ratio is not a spectral dichotomy)**.** *Whenever the denominator is nonzero, define*

$$\Gamma(u) := \frac{\|Q(u,u)\|_2}{\|u\|_2\,\|\operatorname{Def} u\|_2}. \tag{100}$$

Then $\Gamma(u) = 0$ for every non-Killing field in any pressure-linear sector, including every signed curl shell and the entire maximal-torus sector. Therefore $\Gamma \to 0$ does not imply concentration in one Beltrami shell.

Indeed there exist finite sums u_M in the maximal-torus sector with $\Gamma(u_M) = 0$ and with nonzero energy in M distinct Stokes levels. After equalizing the level energies, the Shannon entropy of the coarsened Stokes-level distribution is exactly $\log M$.

Proof. If $Q(u,u) = 0$ and $\operatorname{Def} u \neq 0$, then (100) vanishes. For the second statement choose nonzero toric eigenmodes from M distinct Jacobi levels and rescale them to equal L^2 energies. The toric sector is pressure-linear, so the numerator remains zero. Orthogonality of distinct Stokes eigenspaces gives level probabilities $1/M$ and hence entropy $\log M$. □

Remark XV.4. Thus neither $\Gamma(u) \to 0$ nor a lower bound $\Gamma(u) \ge \gamma_0 > 0$ supplies, by itself, a theorem asserting exactly two possible forms of spectral escape. In particular, entropy divergence still requires a separate energy-spreading hypothesis, and shell-count divergence requires the derivative-budget assumptions stated in the companion note.

The rigidity theorem does not classify arbitrary finite-dimensional reducing spaces outside the Killing-saturated category. The following resonance-slope statement records one precise limitation of a narrower graph ansatz.

A proposed strengthening introduces a family of three-dimensional graph subalgebras $H_c \subset \mathfrak{K}$ and, for a given wave number k, a cancellation condition

$$c = c_*(k) := \frac{k}{k+4}.$$

Even when such a family has been constructed in a specified ansatz, the monotonicity of $c_*(k)$ proves only the following narrow statement.

Proposition XV.5 (Single-slope incompatibility across distinct levels)**.** *Assume a fixed graph subalgebra H_c can preserve a nonzero partial subspace in level k only when $c = c_*(k) = k/(k+4)$. Then the same fixed H_c cannot satisfy that cancellation condition simultaneously for two distinct levels $k_1 \neq k_2$.*

Proof. The function

$$c_*(k) = \frac{k}{k+4}$$

is strictly increasing for $k \geq 0$, since

$$c_*'(k) = \frac{4}{(k+4)^2} > 0.$$

Hence $c_*(k_1) = c_*(k_2)$ implies $k_1 = k_2$. □

Remark XV.6 (Why this is not a global classification theorem). The preceding proposition says nothing by itself about $Q(L_i, L_j)$ for $i \neq j$, about graph subalgebras outside the chosen one-parameter family, about invariant spaces that are not of graph type, or about toric reductions that mix spectral data differently. Therefore monotonicity of $c_*(k)$ cannot imply that *all* finite-dimensional Navier–Stokes reducing spaces are exhausted by a listed collection. The exhaustive statement proved here is instead Theorem IX.2, and it requires the full six-dimensional Killing algebra $\mathfrak{K} \subset W$.

Spectral escape likewise does not force Shannon entropy of the shell-energy distribution to diverge. The following counterexample separates pointwise concentration from energy equipartition.

The projector bounds force rank or bandwidth to grow if a finite $SO(4)$-invariant model is required to capture a fixed fraction of an L^∞ peak. They do *not* force Shannon entropy of the L^2 shell-energy distribution to diverge.

For a nonzero field u, define

$$p_{k,\sigma}(u) = \frac{\|\Pi_{k,\sigma} u\|_2^2}{\|u\|_2^2}, \qquad \mathcal{S}_{\mathrm{spec}}(u) = -\sum_{k,\sigma} p_{k,\sigma} \log p_{k,\sigma}.$$

Theorem XV.7 (Bounded energy and unbounded peak do not force entropy growth)**.** *There exists a sequence of smooth divergence-free fields u_n on S^3_R such that*

$$\|u_n\|_2 = 1, \qquad \|u_n\|_\infty \longrightarrow \infty, \qquad \boxed{\mathcal{S}_{\mathrm{spec}}(u_n) \longrightarrow 0.} \tag{101}$$

Therefore no lower bound of the form

$$\mathcal{S}_{\mathrm{spec}}(u) \gtrsim \log \|u\|_\infty$$

can follow from bounded L^2 energy and spectral projector estimates alone.

Proof. Choose an L^2-unit Killing field K and, for each $n \geq 1$, an L^2-unit extremizer $w_n \in E_{n,+}$ for the sharp projector inequality. Then

$$\|w_n\|_\infty = \sqrt{\frac{(n+1)(n+3)}{3V_R}} \asymp n.$$

Let $\varepsilon_n = n^{-1}$ and define

$$u_n = \sqrt{1-\varepsilon_n}\, K + \sqrt{\varepsilon_n}\, w_n.$$

The two terms are L^2-orthogonal, so $\|u_n\|_2 = 1$. By the reverse triangle inequality,

$$\|u_n\|_\infty \geq \sqrt{\varepsilon_n}\|w_n\|_\infty - \|K\|_\infty \asymp n^{1/2} - O(1) \to \infty.$$

The shell-energy distribution has only two nonzero probabilities, $1-\varepsilon_n$ and ε_n. Hence

$$\begin{aligned}\mathcal{S}_{\mathrm{spec}}(u_n) &= -(1-\varepsilon_n)\log(1-\varepsilon_n) \\ &\quad - \varepsilon_n \log \varepsilon_n \\ &= O\left(\frac{\log n}{n}\right) \to 0.\end{aligned}$$

□

Remark XV.8. This example also shows why the adaptive rank barrier does not imply a frequency-independent lower bound on the *number* of active shells. A single shell has rank $\asymp k^2$ and can move to arbitrarily high frequency. Thus concentration alone permits a tradeoff between shell count and maximum frequency; packet-specific rates for that tradeoff require additional similarity assumptions and are not part of the present article.

A true entropy statement requires an additional energy-spreading hypothesis.

Proposition XV.9 (Conditional entropy lower bound)**.** *Suppose a set S of M signed shells carries total energy probability $q = \sum_{\alpha \in S} p_\alpha \geq \delta > 0$ and*

$$p_\alpha \leq \frac{C}{M} \qquad (\alpha \in S).$$

Then, whenever $M \geq C$,

$$\boxed{\mathcal{S}_{\mathrm{spec}}(u) \geq \delta \log \frac{M}{C}.} \tag{102}$$

Proof. For $\alpha \in S$, $-\log p_\alpha \geq \log(M/C)$. Therefore

$$\mathcal{S}_{\mathrm{spec}}(u) \geq \sum_{\alpha \in S} -p_\alpha \log p_\alpha \geq q \log(M/C) \geq \delta \log(M/C).$$

□

XVI. SCOPE AND INDEPENDENCE FROM THE SINGULARITY-TRANSFER PROBLEM

Every theorem in the present article is an intrinsic statement about Navier–Stokes geometry, exact reduction, or spectral projection on the round three-sphere. No theorem here assumes the existence of a finite-time singular similarity packet.

The motivating Zenodo preprint of Batenin and Preece [1] additionally formulates a conditional transfer of a forced singular branch to the sphere. That transfer construction and its packet-specialized analytic machinery are outside the scope of this article. In particular, none of the Killing–Beltrami interaction, connection-curvature, rigidity, Casimir, toric, projector, Bessel-potential, energy, or helicity results depends on the inner-solver, renormalized matching, or annular-cycle stability hypotheses of the transfer construction.

XVII. CONCLUSION

The central structure on the round three-sphere is the exact chiral Killing–Beltrami intertwining law. It organizes shellwise skew dynamics, a canonical nonflat connection, exact viscous reductions, a converse rigidity theorem for Killing-saturated spaces, and the Casimir representation of Stokes dissipation.

This organization also separates the present contribution from established results. Conservation of the Killing component belongs to the existing geometric Navier–Stokes theory [5], and exact nonstationary Euler solutions on the round three-sphere have independent precedents [7]. The distinct contribution here is the full chiral Killing–Beltrami interaction law and the rigidity, connection, and Casimir structures that follow from it.

Compact homogeneity adds the sharp projector theory and quantitative spectral concentration bounds. These statements are intrinsic. Any specialization to the relative singularity construction of Ref. [1] requires additional packet assumptions and is not asserted here.

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OpenAI ChatGPT was used substantively to test candidate derivations, audit logical dependencies, organize literature, and assist with LaTeX preparation. The tool was directed through explicit mathematical prompts and instructed to distinguish proved statements from conditional or conjectural extensions. Retained outputs were checked against explicit derivations, the Zenodo source preprint, and the cited literature. The human authors remain responsible for final verification and for all scientific content of the submitted manuscript.

DATA AVAILABILITY

This is a purely mathematical work and no data were created or analyzed in this study.

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